

Module 01

Interação Pessoa-Máquina

(IPM) · Human-Computer Interaction

Álvaro Figueira (arfiguei@fc.up.pt) · Diogo Abrantes

1.º semestre 2026/27 · DCC · FCUP

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Overview of course contents

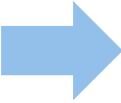
Fundamentals of HCI

(“Interação Pessoa-Máquina” / “IPM”)

Theory

The process to design interfaces

Usability



Implementation of Interactive Systems

Prototyping

Interactive technologies

Application and validation of the methodologies used

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Main topics to be covered



Concepts of the design process



Model of the Human Processor

Perception, attention, memory



Prototyping

Low to high fidelity, tools



Usability engineering

Problem definition, development cycle, evaluation



Interaction as sentiment



History of HCI / IPM

(making it short!)

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Some references (optional)



Sketching User Experiences: The Workbook

Saul Greenberg, Sheelagh Carpendale, Nicolai Marquardt, Bill Buxton

Morgan Kaufmann (2012)



About Face: The Essentials of Interaction Design

Alan Cooper, Robert Reimann, David Cronin, Christopher Noessel

4th edition (2014). The 3rd edition exists online as a PDF



Interaction Design: Beyond Human-Computer Interaction

Jenny Preece, Helen Sharp, Yvonne Rogers

5th edition (2019), 6th edition (2023)



Measuring the User Experience

Tom Tullis, Bill Albert

3rd edition (2022)

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Course frequency



Attending lectures

- Theoretical: presential
- Lab: presential (work in group)
- Important to attend the theoretical sessions as well



Lab sessions

- At least 70% attendance in lab sessions
- **3 misses at most**



All coursework

- You need to **complete all coursework**
- Two group reports, one prototype, two presentations

Assessment and evaluation

- **Based on 3 components:**
 - The First Report: group work (**R1**), 4 points
 - The Prototype + Second Report: group work (**PR2**), 6 points
 - Final Exam, in Moodle (**FE**), 10 points
- **Final Grade (FG):**
 - $FG = R1 + PR2 + FE$ (and $FE \geq 35\%$)

20 points



Coursework: 10 points (continuous assessment, in group)

Exam: 10 points (individual, minimum 35%)

1 · FIRST REPORT · 4 POINTS

Assessment of the R1 component

-  This coursework is based on the **study and design** of a system capable of performing human-computer interaction
-  The group work involves: research, requirements analysis, implementation methodology and preliminary evaluation results
-  Groups of 4 participants
-  No minimum grade, but the coursework is **continuously assessed**
-  One mid-term presentation of the report (TBC)

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2 · PROTOTYPE + SECOND REPORT · 6 POINTS

Assessment of the PR2 component

-  This work is based on a **prototype implementation**, associated with the previous coursework
-  The group work involves: fast prototyping materials and tools, software use (Figma), and implementation details decided in each case with the teacher
-  The same group as in the previous coursework
-  No minimum grade, but the coursework is continuously assessed
-  One final presentation of the developed prototype + second report

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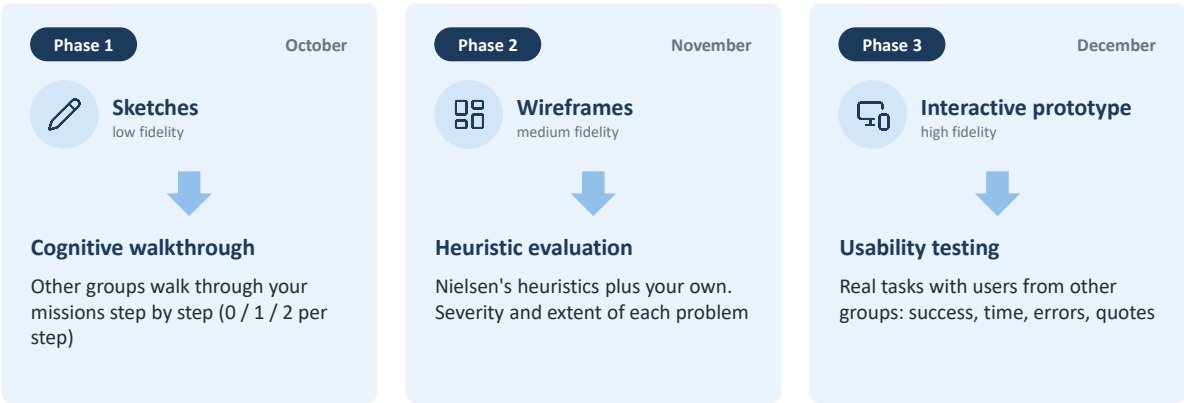
Final exam

- In Moodle, individual
- Multiple choice questions
- Association questions
- Drag and drop questions
- About 70 to 90 minutes duration
- **Minimum: 35% of the exam grade**



The project: one system, three phases, three evaluations

One project for the whole class. Each group builds its own approach and must have something the others don't have: that is the group's gold.



After every phase: evaluation by other groups → written conclusions → the artefact is improved. Exact dates on Moodle.

How the groups work



Groups of 4

Same group for the whole semester (R1 and PR2)



Team leader rotates

A different leader in every session. At the end, the leader says what the group should change to work better



One panel per member

Each member owns one panel of the system, answers for it and presents it



Choose alone, then discuss

Individual choices are made before the group discussion (for example, which features each one sketches)



Evidence in every class

Photos of sketches, votes, matrices and decisions are saved at the end of each session (Moodle)



Evaluation between groups

Missions, evaluator-by-object matrices, independent scores and written conclusions. Evaluators know the domain

Generative AI in this course



Allowed

as instrumental support

- Rephrasing and checking your texts
- Exploring vocabulary and alternatives after your own
- Snippets and help with the prototyping tools
- Summarising references you have read



Not allowed

as a substitute for your thinking

- Generating the sketches or the concepts
- Making the design decisions for the group
- Writing the evaluations or the conclusions
- Producing 'evidence' the group did not create



What counts

only what you own

- What you understood
- What you can explain
- What you validated with users or evaluators
- Every decision must be defended by the group, on request
- Exam also about the coursework

Material you should bring to classes



Sketchbook

A5 or A4



Pen or pencil



Rubber / eraser



Paper for taking notes



A Figma account

Free education plan. We will use it for wireframes and the interactive prototype



Access to Moodle

The group's evidence (photos, matrices, decisions) is uploaded there

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Prototyping

A first example

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Nespresso like a coffee machine



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Nespresso like a coffee machine: what is behind the interface?



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TPC (homework)



1. Take a photo

of some interesting HCI (IPM) system, for good or for bad reasons



2. Annotate the photo

with: your **name**, the **date**, and a **short description** (what it is, why it is interesting)



3. Submit it on Moodle

JPG, less than 1 MB, before the next lecture

Examples



It is a good example, but there is no caption

Examples



Good example

The photo is annotated with the name, the object and the date

